MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, DETAIL LIGHTING FOR DESKS

8 WATTS, 120 VOLTS, 60 HERTZ, SYMBOL 148.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).

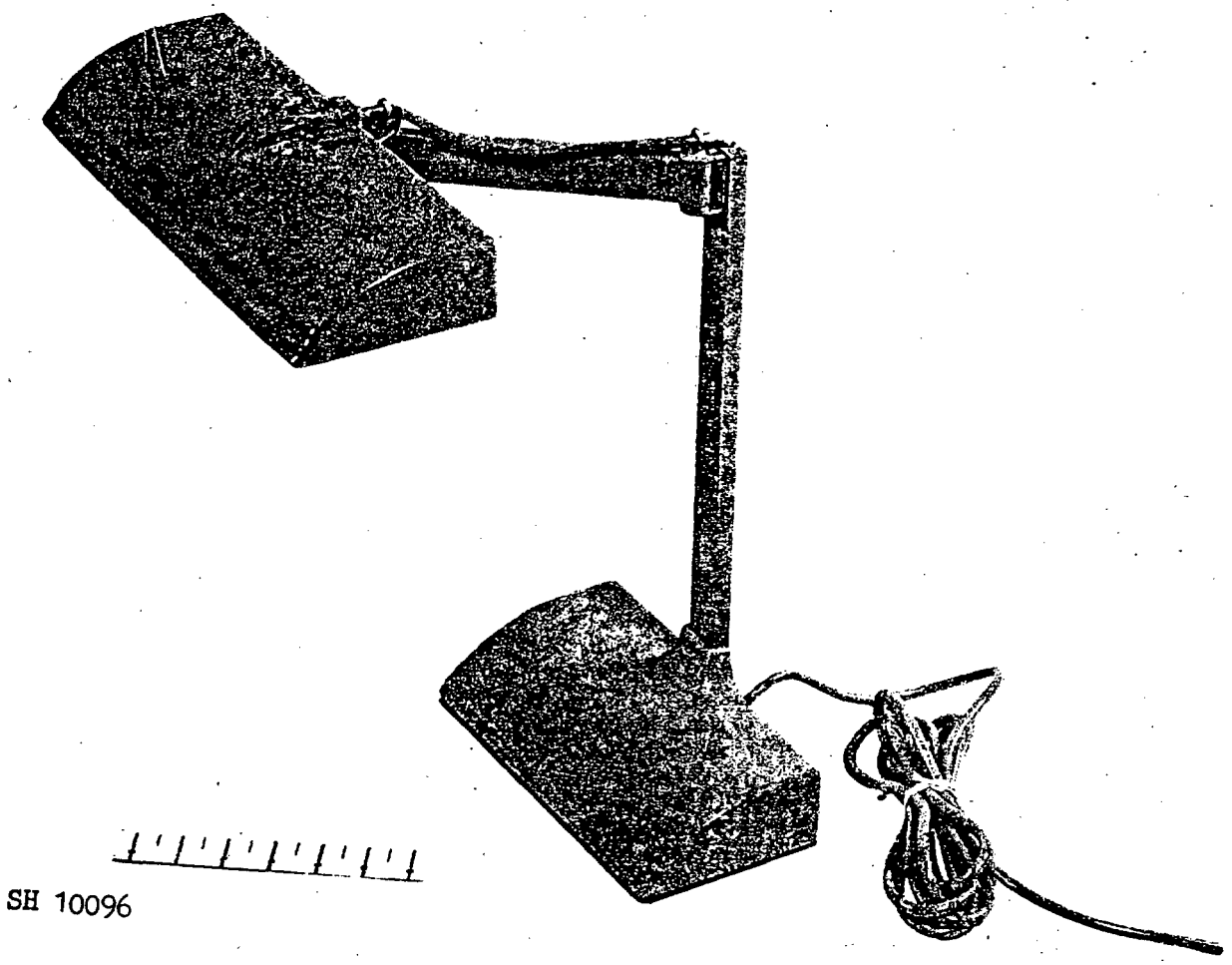
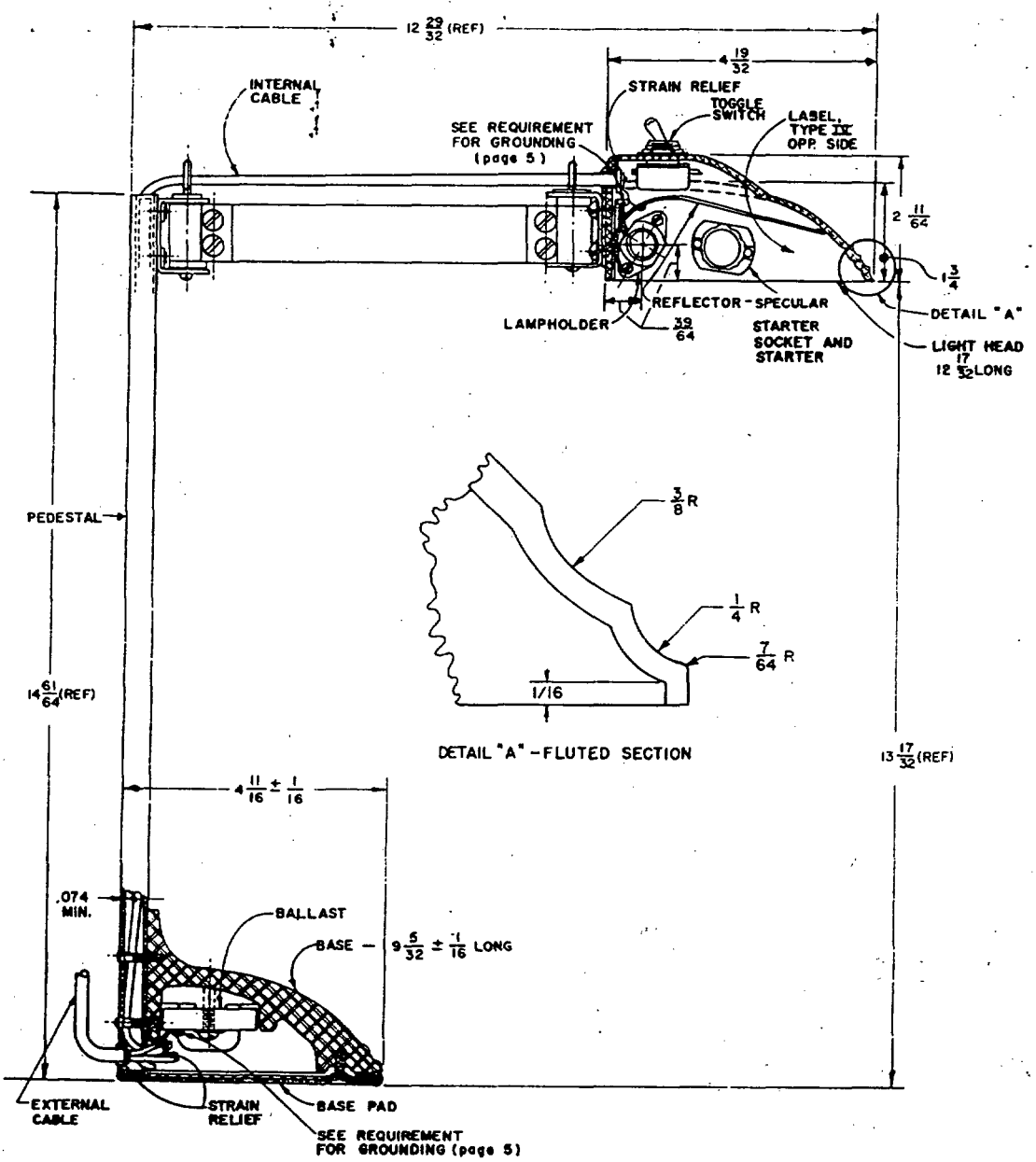


FIGURE 1. Typical view of Symbol 148.2.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6230

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FIGURE 2. Typical section.

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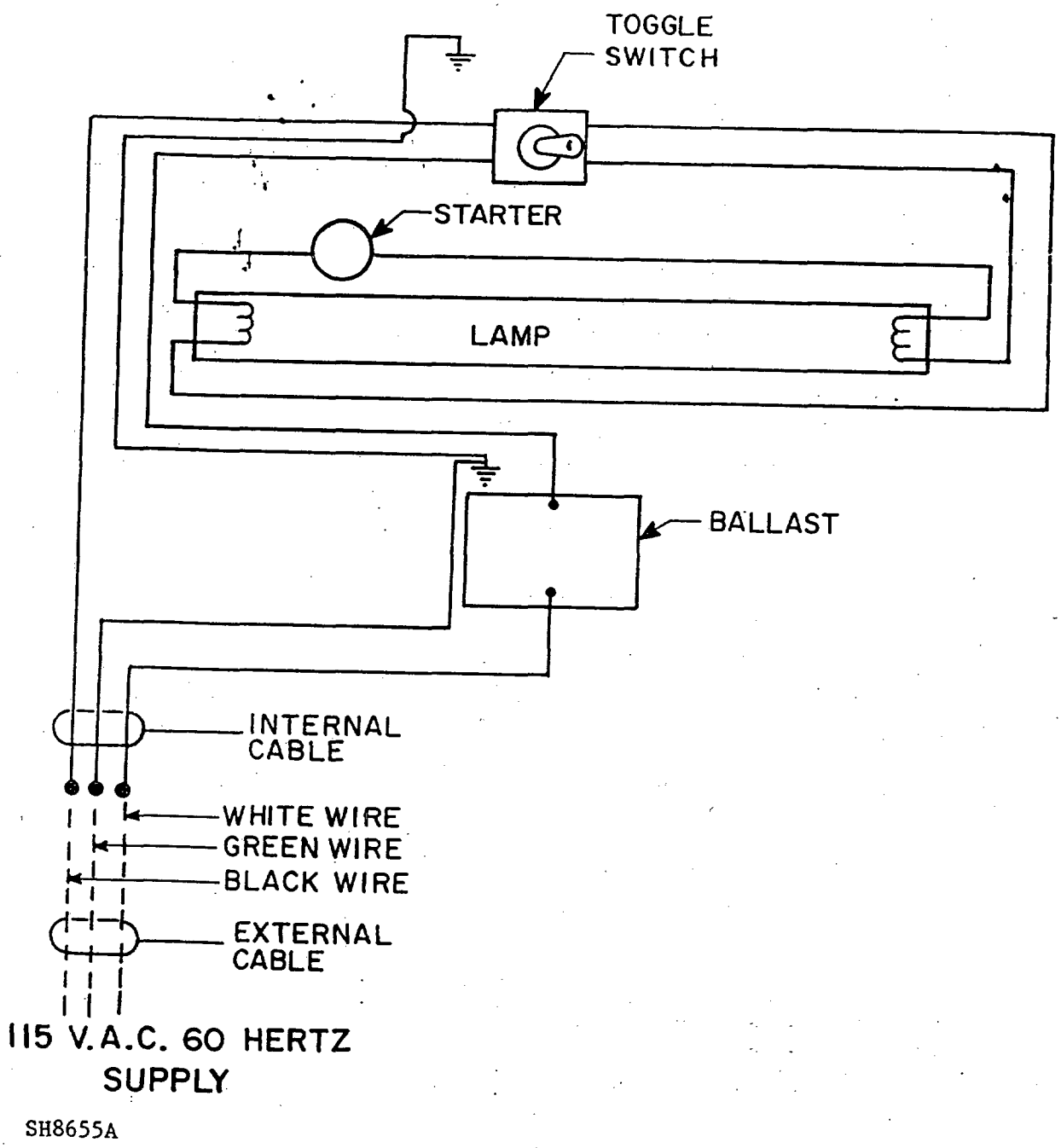
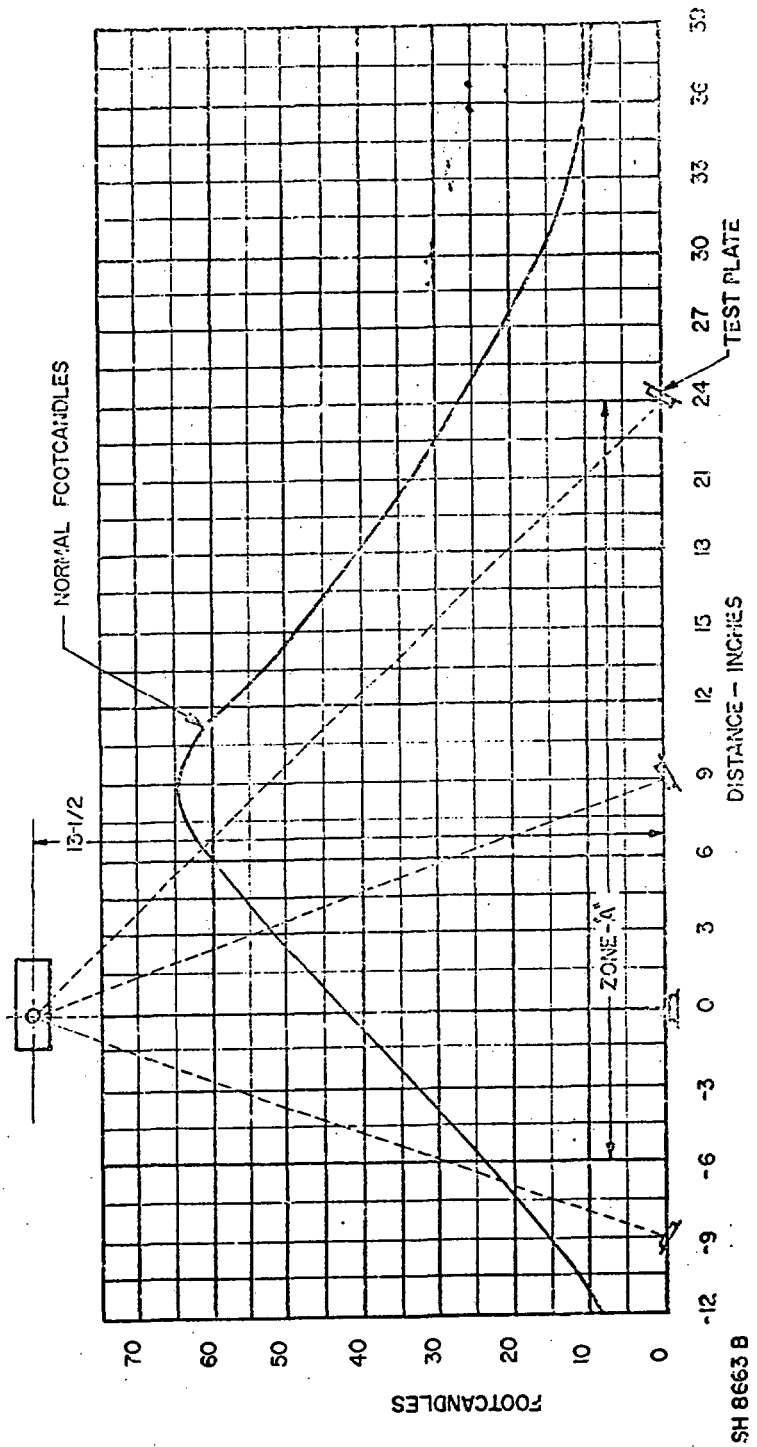


FIGURE 3. Wiring diagram.

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FOOTCANDLE VALUES WITHIN ZONE "A" SHALL BE NOT LESS THAN 90 PERCENT OF THE VALUES SHOWN HEREON. THESE RESULTS SHALL BE MET AFTER CORRECTION TO 380 LUMENS OUTPUT FOR BOTH LAMP AND BALLAST HAS BEEN MADE.

Photometric test data:

Distance: 13½ inches.
Voltage: 115 volts, a.c., 1 phase.
Ambient temperature: 76°F.
Lamp and ballast furnished with fixture shall be used when testing.

FIGURE 4. Footcandle distribution curves.

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REQUIREMENTS:

CLASSIFICATION:
Type: I
Class: 1

SYMBOL NUMBER:
148.2: Light, fluorescent, desk with one 8-watt cool white lamp.

DESIGN:
Electrical rating: 8 watts, 120 volts, 60 Hertz.
Dimensions and configuration: See figures 1 and 2.
Typical section: See figure 2.
Distribution curves: Footcandle (see figure 4).
Cable: Internal and external cable shall be type SJT or SVT construction in accordance with J-C-580 with light gray thermoplastic sheath, except size 18 AWG, 3 conductor. For internal and external cable, the green ground leads inside base cavity shall be terminated with crimp-type terminals and attached under screw and washer. For internal cable, the green ground leads inside the lighting head assembly shall be terminated with crimp-type terminals and connected under round head machine screw and lock-washer. 33 inches of internal, and 10 feet of external cable shall be provided.
Lighting head, supporting arm and base assembly: The lighting head shall be made of 3/32 inch minimum thickness cast aluminum. The supporting arm shall be made of 0.074 minimum thickness extruded aluminum. The lighting head shall be designed to house the necessary parts, and shall be of a shape that will accommodate the filter assembly shown on MIL-F-16377/48, Symbol 121.1 and 121.2. Swivel pins shall be extruded nylon. Swivel brackets and wire loops shall be aluminum. The base shall be cast zinc. Exterior surfaces of the base shall be smooth and free from blemishes. Swivel joints shall hold arm firm at all adjustments. Swivel action shall be smooth for all adjustments.
Wiring diagram: See figure 3.
Reflector: Specular, pre-finished lighting sheet.
Weight: 11 lbs., 10 oz. maximum.
Base pad: Rubber cork, adhesive back, Armstrong Cork Company No. DR-153, or equal. The rubber cork shall be applied to a metal plate for securing to fixture.
Strain relief: Strain relief for cables shall be provided as shown on figure 2.
Designation and marking: The following labels shall be furnished in accordance with MIL-F-16377 applied as shown in figure 2.

(a) One type I.

(b) One type IV. The information to be shown on the type IV label shall be "USE LAMP NSN 6240-00-299-5546 ONLY."

Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein with no deviation.

QUALITY ASSURANCE:

Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection: See table II.
Quality conformance:
Comparison inspection: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation of delivery: See MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

GENERAL INFORMATION:

Assignment of Military part numbers and general information is as follows:

(a) Military part Number: M16377/15-148.2.
(b) National Stock Number: 6230-00-152-2870.
(c) Supersession data : Formerly covered on NAVSHIPS Drawing 9000-S6401-73785 and MS17119.
(d) This light is designed to receive a red and white or blue and white filter shown on MIL-F-16377/48, Symbol 121.1 or 121.2. The filters are not furnished with the light and must be ordered separately when required.

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TABLE I. Standard parts.

Name	Specification or Military part Number	National Stock Number (NSN)	Remarks
Lamp	W-L-116/3	6240-00-299-5546	Industry number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	
Lampholder	MIL-L-970/10	6250-00-377-8590	
Startersocket	MIL-L-970/13	6250-00-295-2738	
Starter	80, type III of W-S-755	6250-00-299-5962	Commercial Designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	
Switch, toggle, DPST	MS16569-1	5930-00-719-9659	
Red and white filter	M16377/48-121.1	6230-00-548-2040	Not furnished
Blue and white filter	M16377/48-121.2		Not furnished

TABLE II. First article and comparison inspection.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Noise	3.5.14	4.8.9	
Grounding circuit	3.6.13.3	4.8.11	
Tilt test:			2/
Light output	3.7.2	4.8.6.2	Footcandle
Magnetic permeability	3.4.8	4.8.18	

1/ Lamp breakage or lamp fall out during this test does not constitute failure.

2/ With the lighting head in any normal position, the light shall sustain a 15 degree tilt in any direction without upsetting.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6230-N266)

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